

Centimeter-Level 3-D Mobile Online Visible Light Positioning System With Single LED Lamp

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Abstract—In this article, we consider a practical indoor 3-D mobile online visible light positioning (VLP) system, where the orientation of the user equipment (UE) is arbitrary. Based on the received signal strength (RSS) of multiple photodetectors (PDs), we formulate the 3-D VLP problem as a nonlinear least squares (NLSs) optimization problem, and then propose a sequential quadratic programming (SQP) positioning algorithm to efficiently calculate UE's location. To obtain more accurate positioning solutions, we further leverage the advantages of deep learning and develop a stochastic gradient descent (SGD)-based VLP algorithm, and achieve an average positioning error of 1.77 cm, which significantly outperforms existing RSS VLP localization methods. Moreover, we design a 3-D mobile online VLP system prototype by using a portable RaspberryPi 4 Model B as the positioning signal processor and data memory, and establish the first publicly available 3-D VLP measured data set, including both RSS and orientation. The proposed positioning schemes are implemented and evaluated via the designed prototype system, which can achieve centimeter-level positioning accuracy (below 1 cm in certain condition).

Index Terms—Arbitrary orientation, received signal strength (RSS), visible light positioning (VLP).

I. INTRODUCTION

THE INDOOR positioning technology has attracted significant attentions due to its key role in numerous

location-aware services, including, but not limited to, indoor navigation, asset tracking, human activity recognition, and intelligent logistics system. By utilizing the widely deployed light emitting diode (LED), visible light positioning (VLP) can simultaneously provide high-accuracy localization and illumination services. Compared with other existing indoor positioning technologies, such as wireless fidelity (WiFi) positioning [1], radio frequency identification (RFID) [2], magnetic information [3], bluetooth [4], and ultrawide-band (UWB) [5], VLP exhibits the advantages of high-positioning accuracy, license-free spectrum, energy efficient, low-multipath effects, high security and no electromagnetic interference, etc. Furthermore, VLP can be widely applied in electromagnetic sensitive areas, where RF radiation is potentially hazardous or even forbidden, e.g., hospitals, nuclear power plants, and mines.

In VLP systems, various techniques can be leveraged for position estimation, such as image sensing [6], [7], received signal strength (RSS) [8], time of arrival (TOA) [9], [10], time difference of arrival (TDOA) [11] and angle of arrival (AOA) [12]. Specifically, the image sensing positioning method requires complicated image sensor hardware (cameras) as the receiver. AOA, TOA and TDOA-based indoor positioning methods require complex signal processing [13], and while RSS is the most commonly used VLP method due to its easy implementation and low-cost characteristics.

According to the required number of lamps, RSS-based VLP technology can be further classified into multilamps positioning schemes [14], [15] and single-lamp positioning schemes [16], where the lamp is utilized as the anchor node for positioning. However, the multilamps positioning schemes in general require at least three anchors for effective localization, which is hard to implement in many practical VLC applications area. For example, in a long corridor or tunnel, the lamps are deployed along a line with large intervals. In these scenarios, multiple lamps may not be visible to the user equipments (UEs) at the same time. To overcome the practical application limitations, the single-lamp VLP schemes were proposed [8], [12], [14], [16], [17]. Specifically, the triangulation positioning methods were developed for 2-D [14] and 3-D [8] VLP, respectively. Using AOA and RSS, a 3-D indoor positioning system was proposed in [12] with the fixed photodetector (PD) orientation. In [16], a 2-D positioning algorithm was presented based on a long short term memory-fully connected network (LSTM-FCN). Fixing the PD orientation, a weighted k -nearest neighbor (KNN) algorithm was applied

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TABLE I
COMPARISON WITH EXISTING POSITIONING WORKS

Work	Method	Areas (L, W, H) m ³	Lamp	2D /3D	Accuracy (cm)	Notes	On/Off line	Receiver Orientation θ	SNR (dB)
[8]	Tri.	(1, 1, 0)	1	2D	2.15	Exp.	Off	0°	/
[18]	Tri.	(0.25, 0.25, 0)	3	2D	1.68	Exp.	Off	0°	/
[7]	Tri.	(1.8, 1.8, 0)	1	2D	2.26	Exp.	On	0°	/
[14]	Tri.	(6, 6, 0)	6	2D	4-6	Exp.	Off	0°	/
[19]	Tri.	(1.5, 1.2, 0)	4	2D	25	Exp.	On	0°	15-60
[20]	Tri.	(0.6, 0.6, 0)	4	2D	4/8	Exp.	On	0°	42-56
[21]	Tri.	(1, 1, 0)	4	2D	1	Sim.	Off	0°	10
[22]	Tri.	(5, 5, 0)	5	2D	3.3	Exp.	On	0°	25
LS	Tri.	(3, 3, 2.2)	1	3D	16.8	Exp.	On	0°	8.5
LS	Tri.	(3, 3, 2.2)	1	3D	26.36	Exp.	On	Arbitrary	8.5
[23]	DL	(1.2, 1.2, 2)	3	3D	11.93	Exp.	Off	0°	/
[24]	DL	(1, 1.1, 2.5)	3	3D	3.65	Exp.	Off	0°	/
[17]	DL	(1.2, 1.2, 0)	4	2D	1	Exp.	Off	0°	/
[15]	DL	(5, 5, 0)	4	2D	2.48	Sim.	Off	0°	/
[16]	DL	(1, 1.1, 0)	1	2D	0.92	Exp.	Off	0°	/
[25]	DL	(0.5, 0.6, 0.8)	3	3D	3.16	Sim.	Off	0°	/
[26]	DL	(1, 1, 0)	3	2D	4.9	Exp.	On	0°	/
[27]	DL	(8, 8, 0)	3	2D	3.4	Sim.	Off	0°	29
[28]	DL	(1, 1, 0)	4	2D	0.7	Exp.	On	0°	/
SGD	DL	(3, 3, 2.2)	1	3D	0.7	Exp.	On	0°	8.5
SGD	DL	(3, 3, 2.2)	1	3D	1.77	Exp.	On	Arbitrary	8.5

in [17] for 2-D fingerprinting localization. The features and the performance of the above works are summarized in Table I. Note that, the existing VLP works only consider 3-D (or 2-D) positioning with upward (or fixed) PD orientation. In practice, the PDs' orientation of mobile users may not be fixed and can be arbitrary.

In this article, for a 3-D VLP system, we develop a least squares (LSs) positioning scheme and a stochastic gradient descent (SGD) positioning scheme for arbitrary UE orientation, as well as the vertical up UE orientation (like most existing works did). More specifically, the main contributions of this work are summarized as follows.

- 1) Based on the RSS at the multi-PDs, we first formulate 3-D VLP problem with arbitrary UE orientation as a nonlinear LS (NLS) optimization problem by exploiting the Lambertian channel model, which is nonconvex and difficult to find the optimal solutions. To overcome this difficulty, we propose a sequential quadratic programming (SQP) positioning algorithm to iteratively solve the NLS problem, where the iteration step sizes are updated according to Wolfe–Powell rule. The proposed SQP scheme can achieve the positioning error of 16.8 cm for the vertical up orientation, and 26.36 cm for the arbitrary orientation.
- 2) Since the 3-D VLP problem is a nonlinear and nonconvex problem, the above proposed SQP algorithms may fall into poor local optimal positioning solutions. To further improve the positioning accuracy, we explore the advantages of deep learning to solve the nonlinear and

nonconvex 3-D VLP problem. More specific, we design SGD-based VLP schemes for the arbitrary and vertical up PDs orientation, respectively. The SGD-based VLP schemes can achieve 0.7 cm for the vertical up PDs orientation, and 1.77 cm for the arbitrary PDs orientation, which can significantly outperform existing RSS VLP localization methods (referring to Table I).

- 3) We further design and implement a 3-D online single-lamp-multi-PDs positioning system prototype. By using a portable RaspberryPi 4 Model B as the positioning signal processor and data memory, the light intensity of multiple PDs and inertial measurement unit (IMU) orientation data are collected simultaneously. Then, we build an RSS and orientation measurements database, which is available at <https://pan.baidu.com/s/1GCvxSaKpnqjd1XqSRsfbfq?pwd=nbjy>. To the best of our knowledge, this is the first publicly available RSS and orientation measured database of 3-D VLP systems, which is free for researchers to perform VLP testing and analysis.

Moreover, we compare the proposed positioning schemes with exiting works in detail from the perspectives of positioning area, number of lamps, positioning dimensions, positioning accuracy, verification methods, processing methods, receiver orientation and SNR, which are listed in Table I. In Table I, “Tri.” means triangulation positioning methods, “DL” means deep learning positioning methods, “Sim.” means simulation results, “Exp.” means experimental results. Specifically, comparing with the existing VLP schemes, the proposed

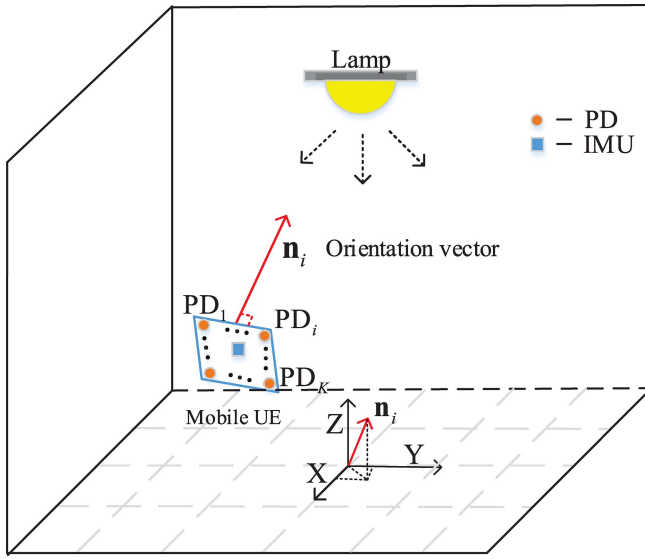


Fig. 1. 3-D VLP system model.

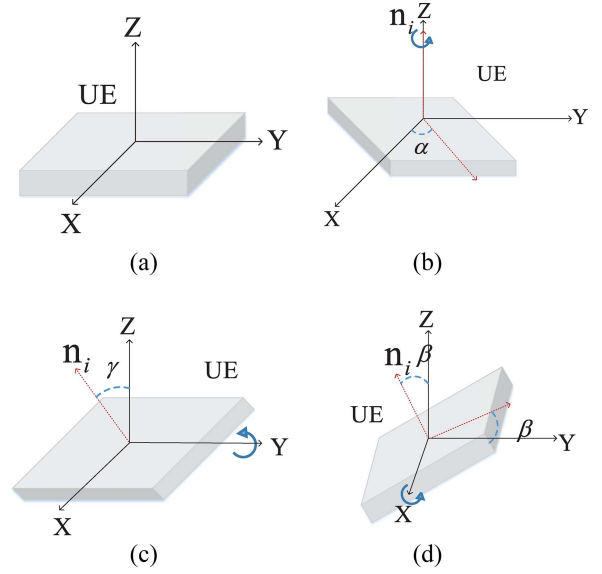
VLP schemes are the first to achieve 3-D positioning with arbitrary receiver orientation. Moreover, comparing with the trilateration-based VLP schemes, positioning areas of the proposed LS schemes are the largest, and the SNRs of the input signals are the lowest. Comparing with the DL-based VLP schemes, the positioning areas of the proposed SGD schemes are the largest, and the SNRs of the input signals of the proposed SGD schemes are the lowest.

The remainder of this article is organized as follows. In Section II, we introduce the 3-D VLP system model. In Section III, we develop 3-D LS-based positioning scheme. In Section IV, we propose the SGD-based positioning scheme. In Section V, we present the 3-D VLP system prototype design and implementation. Section VI evaluates the performance of the proposed positioning algorithms, and finally this article is concluded in Section VII.

II. SYSTEM MODEL

Consider a 3-D single-lamp-multi-PDs positioning system, as shown in Fig. 1, where the lamp with a single LED in the ceiling serves a mobile UE that has a K PDs and an IMU. Let $\mathbf{u}_L = [\vartheta_1, \vartheta_2, \vartheta_3]^T$ and $\mathbf{u}_i \triangleq [u_{i,1}, u_{i,2}, u_{i,3}]^T$ denote the position of the lamp and the i th PD, respectively, where $i \in \{1, 2, \dots, K\}$. Assume that the PDs are placed rigidly on the UE, i.e., all PDs rotate along with the UE. Without loss of generality, assume that the position of the first PD represent the position of the UE. As such, the orientation and the rotation angle of the first PD are the same as those of UE. Moreover, the orientation of the lamp is vertical downward, i.e., $\mathbf{n}_L = [0, 0, -1]^T$.

Based on Euler's rotation theorem [29], any UE's orientation in $\mathbb{R}^3$ space can be uniquely decomposed by three elemental rotations. Fig. 2(a) shows the initial vertical up orientation of UE, and three kinds of rotation are given in Fig. 2(b)–(d), respectively. Specifically, α , β , and γ , respectively, denote rotation angle around the z -axis, x -axis, and y -axis, which

Fig. 2. User receiver orientation diagram. (a) Horizontal. (b) Yaw α . (c) Roll γ . (d) Pitch β .

are called yaw, pitch and roll, respectively. The corresponding rotation matrices $\mathbf{R}_\alpha$, $\mathbf{R}_\beta$, and $\mathbf{R}_\gamma$ are, respectively, given as

$$\mathbf{R}_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1a)$$

$$\mathbf{R}_\beta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta \\ 0 & \sin \beta & \cos \beta \end{bmatrix} \quad (1b)$$

$$\mathbf{R}_\gamma = \begin{bmatrix} \cos \gamma & 0 & \sin \gamma \\ 0 & 1 & 0 \\ -\sin \gamma & 0 & \cos \gamma \end{bmatrix}. \quad (1c)$$

The concatenated rotation matrix can be written as $\mathbf{R} = \mathbf{R}_\alpha \mathbf{R}_\beta \mathbf{R}_\gamma$, which is explicitly given as

$$\mathbf{R} = \begin{bmatrix} \cos \gamma \sin \alpha \sin \beta + \cos \alpha \sin \gamma \\ \sin \alpha \sin \gamma - \cos \alpha \cos \gamma \sin \beta \\ \cos \beta \cos \gamma \end{bmatrix}. \quad (2)$$

Therefore, the orientation of the i th PD can be expressed as $\mathbf{n}_i = \mathbf{R} \mathbf{n}_{0,i}$, where $\mathbf{n}_{0,i} \in \mathbb{R}^{3 \times 1}$ denote the initial orientation of the i th PD. Furthermore, after the rotation, the position of the i th PD is given as

$$\mathbf{u}_i = \mathbf{u}_1 + \mathbf{a}_i \quad (3)$$

where $\mathbf{a}_i = \mathbf{R} \mathbf{a}_{0,i}$ denotes the position bias of the i th PD, $\mathbf{u}_1$ is the position of the first PD, and $\mathbf{a}_{0,i} \in \mathbb{R}^{3 \times 1}$ denotes the position bias relative to the first PD in the initial position.

Let $h_i(t)$ denote the VLC channel between BS and the i th PD at time t , which includes both the line-of-sight (LOS) link and the diffused reflection link. Mathematically, the VLC channel gain $h_i(t)$ is given as [30], [31]

$$h_i(t) = h_{L,i}(t) + h_{d,i}(t - \Delta T_i) \quad (4)$$

where $h_{L,i}$ and $h_{d,i}(t - \Delta T_i)$ denote the gains of the LOS channel and diffused reflection channel, respectively, $\delta(t)$ is the

Dirac function, and ΔT_i denotes the time delay of the non-LOS (NLOS) path. Furthermore, according to Lambertian channel model, the LOS channel gain $h_{L,i}$ is given by [32]

$$h_{L,i} = \frac{(m+1)A_{PD,i}}{2\pi d_i^2} g_f \cos^m(\phi_i) \cos(\psi_i) \text{rect}\left(\frac{\psi_i}{\Psi_c}\right) \quad (5)$$

where $m = -\ln 2 / \ln(\cos \Phi_{1/2})$ denotes the Lambertian emission order, $\Phi_{1/2}$ is the semi-angle at half power. $A_{PD,i}$ represents the receiving area of the i th PD, g_f is the gain of the optical concentrator, d_i denotes the distance between BS and the i th PD, ϕ_i represents the angle between the emitted light and the normal vector of the LED. ψ_i is the angle between the emitted light and the normal vector of the i th PD, and Ψ_c stands for field of view (FOV) of PD. Moreover, $\text{rect}(\psi_i/\Psi_c)$ is given by

$$\text{rect}\left(\frac{\psi_i}{\Psi_c}\right) = \begin{cases} 1, & 0 \leq \psi_i \leq \Psi_c \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where $i \in \mathcal{K}$.

By taking the position $\mathbf{u}_i = \mathbf{u}_1 + \mathbf{a}_i$ and the angle into consideration, we have

$$h_{L,i} = \frac{\gamma_i (\vartheta_3 - \theta_{3,i})^m (\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i)^T \mathbf{n}_i}{\|\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i\|^{m+3} \|\mathbf{n}_i\|} \quad (7)$$

where $\gamma_i \triangleq [(m+1)A_{PD,i}g_f]/(2\pi)$, and $d_i = \|\mathbf{u}_L - (\mathbf{u}_1 + \mathbf{a}_i)\|$.

Furthermore, let $H_i(f)$ denote the frequency domain channel gain of the i th PD $h_i(t)$, which is given by

$$H_i(f) = h_{L,i} + c_{d,i} \quad (8)$$

where $c_{d,i}$ is the power efficiency for the diffuse signal. In addition, $c_{d,i}$ is given by

$$c_{d,i} = \frac{\rho A_{PD,i} e^{j2\pi f \Delta T_i}}{(1-\rho)A_{\text{room}}(1+jf/f_0)} \quad (9)$$

where A_{room} represents the surface area of room, f_0 stands for the cutoff frequency, and ρ is the average reflectivity of the room.

In the VLP system, the transmitted signal of the LED $x(t)$ is given as

$$x(t) = \sqrt{P}s(t) + I_{DC} \quad (10)$$

where P denotes the power gain of the power amplifier of the LED driver, $s(t)$ denotes the united power signal, i.e., $E\{s^2(t)\} = 1$, and I_{DC} stands for the DC bias. Thus, the received signal of the i th PD $y_i(t)$ is given as

$$y_i(t) = h_i(t) * x(t) + z_i(t) \quad (11)$$

where $z_i(t) \sim \mathcal{N}(0, \sigma_i^2)$ denotes the additive white Gaussian noise (AWGN). Therefore, the received power of the i -PD is given as

$$P_{y,i} = \frac{1}{T} \int_0^T (h_{L,i} + c_{d,i})^2 x(t)^2 dt + \sigma_i^2 \quad (12a)$$

$$= (h_{L,i}^2 + c_{d,i}^2 + 2h_{L,i}c_{d,i})P_s + \sigma_i^2 \quad (12b)$$

where T represents the symbol period of the transmitted signal.

III. NONLINEAR LEAST SQUARES POSITIONING SCHEMES

In this section, we investigate the NLS positioning scheme by taking the arbitrary PDs orientation into consideration, where the orientation information is obtained from the IMU at the UE.

Since the received power of the PD is a function of UE's location, the location can be jointly calculated based on the received power of multi-PDs. Specifically, the received power of PDs $\{P_{y,i}\}_{i=1}^K$ in (12b) can be reformulated as

$$h_{L,i} = \sqrt{\frac{P_{y,i} - \sigma_i^2}{P_s}} - c_{d,i}. \quad (13)$$

Combining (7) and (13), we have

$$\sqrt{\frac{P_{y,i} - \sigma_i^2}{P_s}} - c_{d,i} = \frac{\gamma_i (\vartheta_3 - \theta_{3,i})^m (\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i)^T \mathbf{n}_i}{\|\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i\|^{m+3} \|\mathbf{n}_i\|}. \quad (14)$$

Since there are K PDs at UE, the UE's position $\mathbf{u}_1$ can be obtained by solving following:

$$\begin{cases} \sqrt{\frac{P_{y,1} - \sigma_1^2}{P_s}} - c_{d,1} = \frac{\gamma_1 (\vartheta_3 - \theta_{3,1})^m (\mathbf{u}_L - \mathbf{u}_1)^T \mathbf{n}_1}{\|\mathbf{u}_L - \mathbf{u}_1\|^{m+3} \|\mathbf{n}_1\|} \\ \vdots \\ \sqrt{\frac{P_{y,K} - \sigma_K^2}{P_s}} - c_{d,K} = \frac{\gamma_K (\vartheta_3 - \theta_{3,K})^m (\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_K)^T \mathbf{n}_K}{\|\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_K\|^{m+3} \|\mathbf{n}_K\|} \end{cases} \quad (15)$$

Note that, there are $(K+3)$ variables in (15), i.e., $\{c_{d,i}\}_{i=1}^K$ and $\mathbf{u}_1$, which is an under-determined equation with infinite solutions. Since the values of NLOS terms $\{c_{d,i}\}_{i=1}^K$ are significantly lower than that of LOS term $h_{L,i}$, we approximate the values of $\{c_{d,i}\}_{i=1}^K$ to be the same, i.e., $c_{d,i} = c_d \forall i$. Thus, (15) can be re-expressed as

$$\begin{cases} \sqrt{\frac{P_{y,1} - \sigma_1^2}{P_s}} - c_d = \frac{\gamma_1 (\vartheta_3 - \theta_{3,1})^m (\mathbf{u}_L - \mathbf{u}_1)^T \mathbf{n}_1}{\|\mathbf{u}_L - \mathbf{u}_1\|^{m+3} \|\mathbf{n}_1\|} \\ \vdots \\ \sqrt{\frac{P_{y,K} - \sigma_K^2}{P_s}} - c_d = \frac{\gamma_K (\vartheta_3 - \theta_{3,K})^m (\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_K)^T \mathbf{n}_K}{\|\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_K\|^{m+3} \|\mathbf{n}_K\|} \end{cases} \quad (16)$$

For convenience, we introduce the following variables:

$$r_i(\mathbf{u}_1) \triangleq \frac{\sqrt{P_{y,i} - \sigma_i^2}}{P_s} - c_d - \frac{\gamma_i (\vartheta_3 - \theta_{3,i})^m (\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i)^T \mathbf{n}_i}{\|\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i\|^{m+3} \|\mathbf{n}_i\|} \quad (17)$$

$$\mathbf{r}(\mathbf{u}_1) \triangleq [r_1(\mathbf{u}_1), r_2(\mathbf{u}_1), \dots, r_K(\mathbf{u}_1)]^T \quad (18)$$

$$f(\mathbf{u}_1) \triangleq \frac{1}{2} \|\mathbf{r}(\mathbf{u}_1)\|^2. \quad (19)$$

Then, (16) can be equivalently formulated as a NLS optimization problem given by

$$\min_{\mathbf{u}_1} f(\mathbf{u}_1) = \frac{1}{2} \|\mathbf{r}(\mathbf{u}_1)\|^2 \quad (20)$$

which is nonconvex and difficult to find the optimal solutions. To overcome this difficulty, we exploit the SQP algorithm [33] to iteratively solve the NLS problem. Specifically, the $(i+1)$ th iteration point $\mathbf{u}_1^{[i+1]}$ is updated as

$$\mathbf{u}_1^{[i+1]} = \mathbf{u}_1^{[i]} + \mathbf{d}^{[i]} \quad (21)$$

Algorithm 1 SQP Positioning Algorithm

- 1: Initialize $\mathbf{u}_1^{[0]}$, step-size $\alpha_0 = 1$ and the convergence tolerance $0 < \mu < 1$;
- 2: **repeat**
- 3: Calculate $\mathbf{r}(\mathbf{u}_1^{[i]})$ and the Hessian matrix $H(\mathbf{u}_1^{[i]})$;
- 4: Update the descending direction $\eta^{[i]}$ in (22);
- 5: Calculate the step-size α_i by Wolfe-Powell rule in (24);
- 6: Update $\mathbf{u}_1^{[i+1]} = \mathbf{u}_1^{[i]} + \alpha_i \eta^{[i]}$;
- 7: $i \leftarrow i + 1$;
- 8: **until** $\|\mathbf{u}_1^{[i-1]} - \mathbf{u}_1^{[i]}\| \leq \mu$;
- 9: Output the location solution $\mathbf{u}_1^{[i+1]}$.

where $\mathbf{d}^{[i]} = \alpha_i \eta^{[i]}$ represents the descending direction vector corresponding to the i th iteration, α_i denotes the stepsize, and $\eta^{[i]} \in \mathbb{R}^{3 \times 1}$ denotes the descending direction. The optimal descending direction $\eta^{[i]}$ satisfies the following:

$$H(\mathbf{u}_1^{[i]}) \eta^{[i]} = -\nabla f(\mathbf{u}_1^{[i]}) \quad (22)$$

where $\nabla f(\mathbf{u}_1^{[i]})$ and $H(\mathbf{u}_1^{[i]})$ denote the gradient and the Hessian matrix of $f(\mathbf{u}_1^{[i]})$, respectively. Furthermore, $H(\mathbf{u}_1^{[i]})$ is given as

$$H(\mathbf{u}_1^{[i]}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_1^2} & \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_1 \partial u_2} & \cdots & \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_1 \partial u_k} \\ \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_2 \partial u_1} & \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_2^2} & \cdots & \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_2 \partial u_k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_k \partial u_1} & \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_k \partial u_2} & \cdots & \frac{\partial^2 f(\mathbf{u}_1^{[i]})}{\partial u_k^2} \end{bmatrix}. \quad (23)$$

To choose a proper step size with a sufficient decrease, the step size α_i is calculated according to the Wolfe-Powell rule, i.e.,

$$f(\mathbf{u}_1^{[i]} + \alpha_i \eta^{[i]}) \leq f(\mathbf{u}_1^{[i]}) + \rho \alpha_i \nabla f(\mathbf{u}_1^{[i]})^T \eta^{[i]} \quad (24a)$$

$$\nabla f(\mathbf{u}_1^{[i]} + \alpha_i \eta^{[i]})^T \eta^{[i]} \geq \xi \nabla f(\mathbf{u}_1^{[i]})^T \eta^{[i]}, \xi \in (\rho, 1) \quad (24b)$$

where $\rho \in (0, 0.5)$ and $\xi \in (\rho, 1)$ are given parameters. In summary, the proposed SQP positioning algorithm is presented in Algorithm 1.

For some practical VLP applications, the orientation of PDs may keep vertical up, such as robot cars, cargo tags, which is a special case of arbitrary PDs orientation. In this special case, $\mathbf{n}_i = \mathbf{e}_3 = [0, 0, 1]^T \forall i = 1, \dots, K$, and the (15) can be further simplified as

$$h_{L,i} = \frac{\gamma_i (\vartheta_3 - \theta_{3,i})^m (\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i)^T \mathbf{e}_3}{\|\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i\|^{m+3}}. \quad (25)$$

Furthermore, the objective function of the NLS problem (20) can be rewrote as

$$r_i(\mathbf{u}_1) = \frac{\sqrt{P_{y,i} - \sigma_i^2}}{P_s} - c_d - \frac{\gamma_i (\vartheta_3 - \theta_{3,i})^m (\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i)^T \mathbf{e}_3}{\|\mathbf{u}_L - \mathbf{u}_1 - \mathbf{a}_i\|^{m+3}}. \quad (26)$$

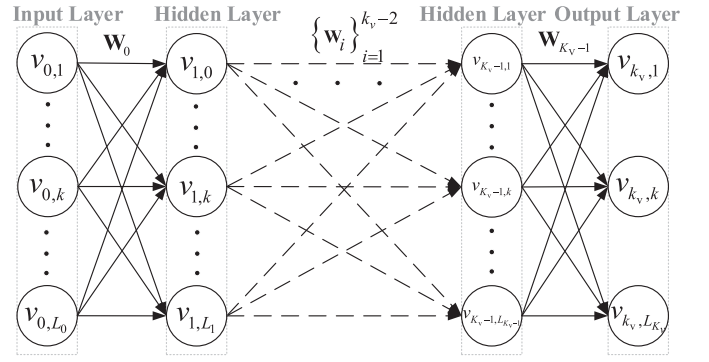


Fig. 3. Proposed SGD positioning network.

Therefore, the proposed SQP positioning Algorithm 1 can also be applied. The performance of Algorithm 1 will be further evaluated in Section VI.

IV. DEEP LEARNING POSITIONING SCHEMES

According to (14), the received power $\{P_{y,i}\}_{i=1}^{K_v}$ are non-linear and nonconvex functions of the localization $\mathbf{u}_1$. Therefore, there are many local optimal localization solutions for the 3-D VLP problem (16). The proposed SQP algorithm may fall into a local optimal solution during the iterative process, which may lead to high-positioning errors. Given the above issues, we turn to apply the data-driven method. Particularly, deep learning has flexible and powerful processing capabilities for complex nonlinear and nonconvex optimization problem. Here, we propose a SGD-based deep learning positioning network to solve the 3-D VLP problem.

Specifically, as shown in Fig. 3, the proposed 3D-VLP SGD network includes K_v layers, i.e., $\{\mathbf{v}_i\}_{i=1}^{K_v}$, one input layer $\mathbf{v}_0 \in \mathbb{R}^{K+3}$, K_v hidden layers $\{\mathbf{v}_i \in \mathbb{R}^{L_i}\}_{i=1}^{K_v-2}$, and one output layer $\mathbf{v}_{K_v} \in \mathbb{R}^3$. Moreover, let $\mathbf{W}_i = [\mathbf{w}_1^{[i]}, \dots, \mathbf{w}_{L_i}^{[i]}] \in \mathbb{R}^{L_{i+1} \times L_i}$ denote the weight matrix between the i th layer and the $(i+1)$ th layer, where $\mathbf{w}_k^{[i]} = [w_{k,1}^{[i]}, \dots, w_{k,L_{i+1}}^{[i]}]^T \in \mathbb{R}^{L_{i+1}}$ denote the weight vector between the i th layer and the nodes of the $(i+1)$ th layer for the k th node.

The SGD positioning network is trained by multiple small batch samples. Assume there are N_b sampled data vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_{N_b}\}$ for each batch, where $\mathbf{b}_i = [P_{y,1}, \dots, P_{y,K}, \mathbf{n}_1^T]^T$ denote the i th sample vector of the received power of K PDs and the orientation of the first PD. Moreover, let $\mathbf{u}_1^{(i)}$ denote the corresponding position of the i th sample vector $\mathbf{b}_i$. For the SGD-based 3D-VLP network, we first assign the sampled data vectors to the input layer node in sequence, i.e., $\mathbf{v}_0 = \mathbf{b}_i$, where $i = 1, \dots, N_b$. Then, the values of the nodes in the i th layer is updated using rectified linear unit (ReLU) activate function, which is given as

$$\mathbf{v}_k = \max(\mathbf{0}, \mathbf{W}_{k-1} \mathbf{v}_{k-1}), \quad k = 1, \dots, N_b \quad (27)$$

where $\max(\mathbf{x}, \mathbf{y}) = \{\max(x_i, y_i)\}$ denote the activation function.

Then, the output layer predicts the 3-D position vector of the first PD, i.e., $\mathbf{v}_{K_v} \triangleq [\hat{u}_1, \hat{u}_2, \hat{u}_3]$. For the i th sample vector $\mathbf{b}_i$,

Algorithm 2 SGD Positioning Algorithm

Input: Set the learning rate α , the stop condition parameter $\delta = 1.0 \times 10^{-10}$, number of sample batches N_b , and $i = 0$;

- 1: **while** $J(\{\mathbf{W}_k\}_{k=0}^{K_v-1}) \geq \delta$ **do**
- 2: **while** $i \leq N_b$ **do**
- 3: Select a batch of samples from the dataset and calculate the value of $L^{(i)}(\mathbf{v}_{K_v}, \mathbf{u}_1^{(i)}, \{\mathbf{W}_k\}_{k=0}^{K_v-1})$;
- 4: $i = i + 1$;
- 5: **end while**
- 6: Calculate $J(\{\mathbf{W}_k\}_{k=0}^{K_v-1})$ in (29);
- 7: Update $\{\mathbf{W}_l\}_{l=0}^{K_v-1}$ according to (30);
- 8: **end while**

Output: $\mathbf{v}_{K_v}$.

let $L^{(i)}$ denote the loss function between the predicted position vector $\mathbf{v}_{K_v}$ and the exact position vector (label) $\mathbf{u}_1^{(i)}$, i.e.,

$$L^{(i)}(\mathbf{v}_{K_v}, \mathbf{u}_1^{(i)}, \{\mathbf{W}_k\}_{k=0}^{K_v-1}) = \|\mathbf{v}_{K_v} - \mathbf{u}_1^{(i)}\|_2. \quad (28)$$

Therefore, for each batch, the mean loss of the N_b sampled vectors is a function of the weight matrix, i.e.,

$$J(\{\mathbf{W}_k\}_{k=0}^{K_v-1}) = \frac{1}{N_b} \sum_{i=1}^{N_b} L^{(i)}(\mathbf{v}_{K_v}, \mathbf{u}_1^{(i)}, \{\mathbf{W}_k\}_{k=0}^{K_v-1}). \quad (29)$$

Then, based on the back propagation, the weight matrix $\mathbf{W}$ is updated by

$$\mathbf{W}_l \leftarrow \mathbf{W}_l - \alpha \frac{\partial J(\{\mathbf{W}_k\}_{k=0}^{K_v-1})}{\partial \mathbf{W}_l}, l = 0, \dots, K_v - 1 \quad (30)$$

where α denotes the step size. In summary, the proposed SGD positioning algorithm is listed in Algorithm 2.

When the orientation of PDs is vertical up, the proposed SGD network can be simplified to a lightweight SGD network without IMU input, which requires less numbers of layers and less calculation time. Specifically, a small batch of SGD network is designed for processing, and each batch $\{\mathbf{b}_1, \dots, \mathbf{b}_{N_b}\}$ includes N_b sampled data vectors, where $\mathbf{b}_i = [P_{y,1}, \dots, P_{y,K}]^T$ denotes the received power of the i th sample vector of K PDs. Compared with the previous arbitrary case, the inputs of the network is reduced to received power of K PDs, i.e., $\{P_{y,i}\}_{i=1}^K$, and the number of hidden layers is also significantly reduced. In this way, based on the K PDs input data vectors, we may obtain the position of the UE-based the simplified Algorithm 2, whose details are omitted for brevity.

V. 3-D VLP PROTOTYPE DESIGN AND IMPLEMENTATION

In this section, we introduce the hardware platform design of 3-D online VLP system, which can be used to implement the proposed positioning schemes. In the 3-D VLP system prototype, the transmitter includes a commercial single LED luminaire and the LED drive control circuit that can adjust the brightness of the light. Note that, for the illumination consideration, multiple LEDs can be integrated in the luminaire as a single anchor node for positioning.

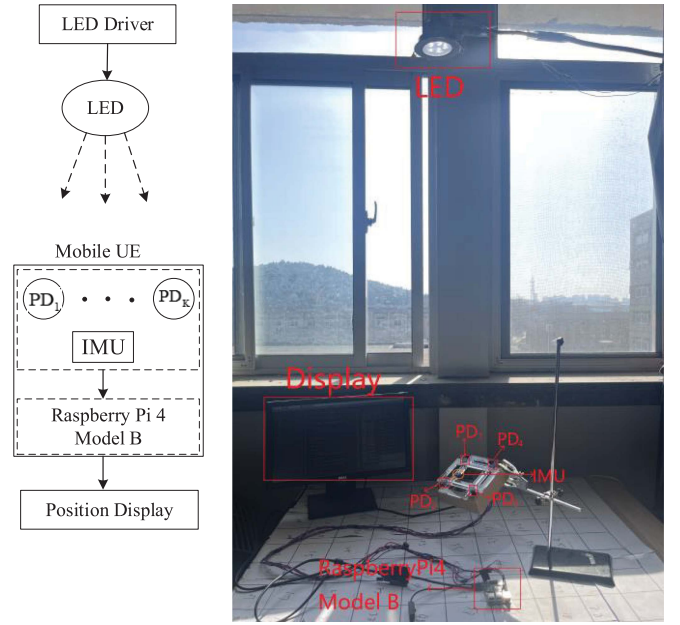


Fig. 4. Proposed 3-D VLP prototype.

At the VLP receiver side, there are four PDs ($K = 4$) to convert optical signals into electrical signals, and an IMU to measure the real-time orientation of the receiver. The receiver is connected to a portable RaspberryPi 4 Model B processor for data collection and online positioning algorithms processing, and one display is to present the real-time positioning results.

In practice, the received signals may be influenced by the background light and interference. Thus, we further introduce a background noise filtering (BNF) algorithm, which can automatically detect and eliminate the background light to reduce the impact of background noise. The BNF algorithm can significantly improve the robustness of the proposed VLP schemes. The system structure and the prototype are shown in Fig. 4. Our proposed 3-D online VLP system prototype can achieve high-accuracy positioning in a bright daylight environment.

Training Set Collection: First, the light intensity data collected by PD and the angle data measured by IMU are synchronized through the our designed software program. Then, in a 3-D space of $(300 \times 300 \times 220)$ cm³, each interval of 20 cm in the horizontal and vertical directions is a data sampling position point, with a total of 2250 data sampling position point. Each point collects 20 different angles and corresponding light intensities, and there are 45000 sampled training data.

Based on the received light intensity and orientation information, the online positioning schemes are implemented in Python language and executed on the RaspberryPi 4 Model B processor. The detailed parameters of the prototype are provided in Table II. Note that, the data sampling, data training, data preprocessing, and the proposed online positioning schemes are all implemented in the proposed positioning prototype system, and the evaluation will be presented in the next section.

TABLE II
HARDWARE PARAMETERS OF 3-D ONLINE VLP SYSTEM PROTOTYPE

System modules	
Processor	RaspberryPi4 model B
Positioning range(L × W × H)	(300 × 300 × 220)cm ³
LED	5W-LED
TX Driver	9018 Triode
RX Photodiode	TS2516
RX Angle sensor	MPU6050
TX characteristic	
Light Power LED	(4 – 7)W
LED FoV	120°
Illuminance range	> 1Lux
RX characteristic	
Lambert Coefficient	$m = 1$
photodiode FoV	90°
Ψ_c	90°
Photodiode gain A_{PD}	1
Photodiode receiving area	1cm ²

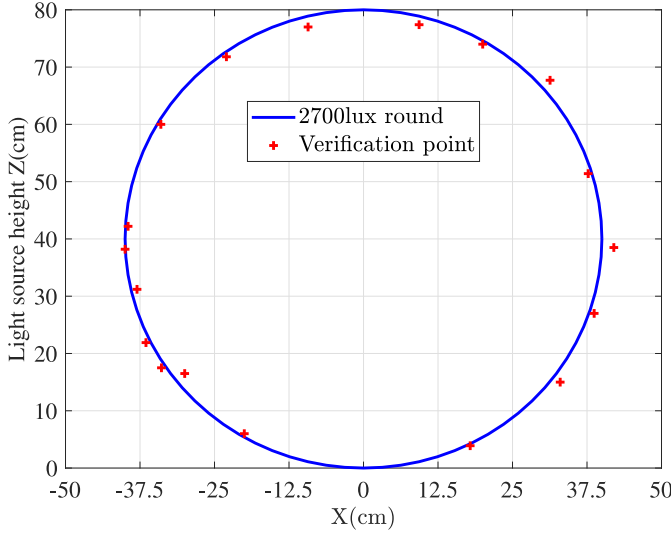


Fig. 5. 2-D iso-illuminance fitting curve of LED with 2700 lux.

VI. EXPERIMENTAL VERIFICATION AND THEORETICAL RESULTS

In this section, we evaluate the proposed 3-D online VLP schemes via our VLP prototype system. As shown in Fig. 4, our prototype is tested in the laboratory near the window with in the (1.5 × 1.5 × 1.5) m² test area. The maximum vertical distance from the LED to PDs is 2 m.

A. LS Positioning Schemes

To verify the accuracy of lambert model, we first test the iso-illuminance spherical surface of the LED. Fig. 5 shows the 2-D equal illumination fitting curve of LED with 2700 lux. It can be seen from Fig. 5 that the test points with the same illuminance can be approximately fitted into a circle, which verifies the accuracy of the LED spherical lambert radiation model.

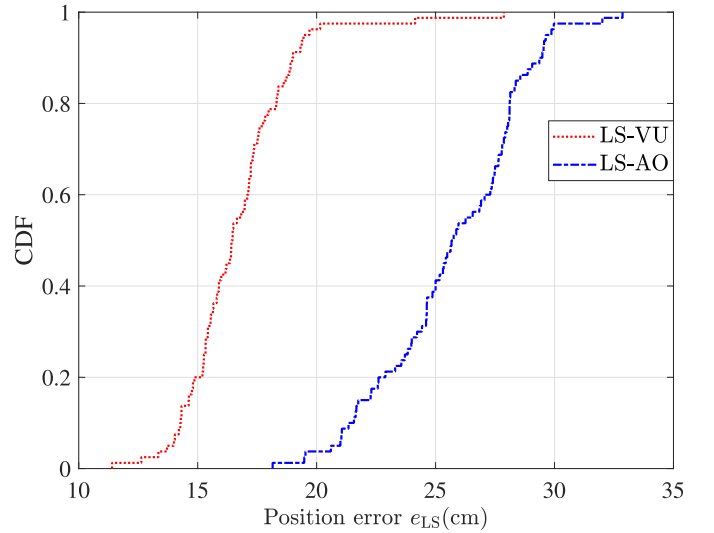


Fig. 6. Positioning errors CDF of the LS VLP schemes.

Let LS-VU denote the LS scheme with vertical up UE orientation, and LS-AO denote the LS scheme with arbitrary UE orientation. Fig. 6 compares the positioning errors CDF of LS-VU and LS-AO schemes. One can observe that the positioning error of LS-VU scheme is less than that of LS-AO scheme. We conclude that the vertical up orientation can contribute to the positioning accuracy. More specifically, the average positioning errors of LS-VU and LS-AO schemes are 16.8 and 26.3 cm, respectively.

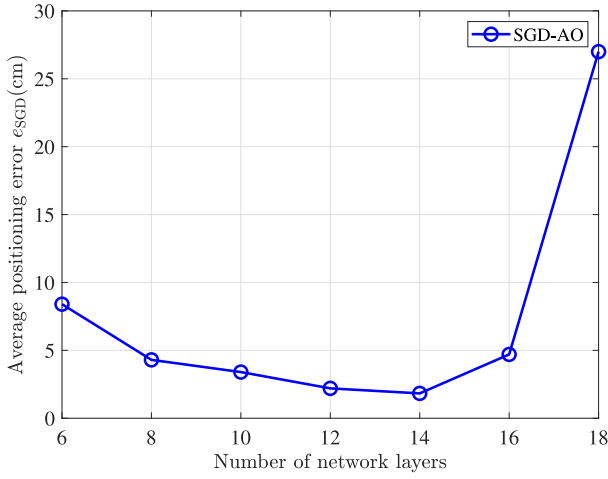
B. SGD Positioning Schemes

In this section, SGD-AO denotes the SGD scheme with arbitrary UE orientation. Since the performance of deep learning does not increase monotonically with the number of network layers, we first determine the optimal number of layers of the SGD-AO VLP network.

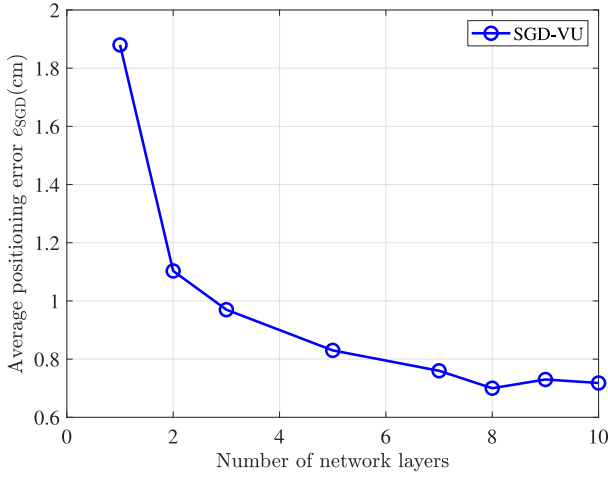
Fig. 7(a) shows the positioning errors versus the number of layers of the SGD-AO network. It can be seen that the positioning error first decreases and then increases as the number of network layers gets large. We may observe that the optimal number of network layers corresponding to the lowest positioning error is 14. Fig. 7(b) shows the positioning errors versus the number of layers of the SGD-VU (SGD with vertical up UE orientation) network. It can be seen that the positioning errors first decreases and then maintains a short fluctuation with the increase of network layers, and has the best positioning accuracy when the number of network layers is 8.

Fig. 8(a) and (b) depict the positioning errors of SGD-VU network and SGD-AO network versus the number of iterations. It can be seen that the positioning error first declines rapidly, and then tends to be constant as the number of iterations increases. After 600 iterations, the network positioning error converges.

Fig. 9 compares positioning errors CDF of SGD-VU and SGD-AO schemes. It can be seen that the positioning error of SGD-VU network is less than that of SGD-AO network. The average positioning errors of LS-VU and LS-AO schemes



(a)



(b)

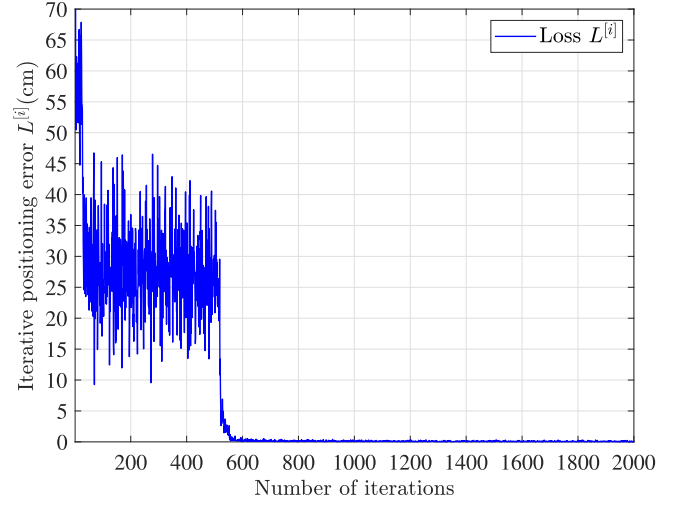
Fig. 7. (a) Positioning errors versus the number of layers of the SGD-AO network. (b) Positioning errors versus the number of layers of the SGD-VU network.

are 0.7 and 1.77 cm, respectively. Compared with Fig. 6, the average positioning errors of SGD-VU and SGD-AO are significantly lower than those of LS-VU and LS-AO schemes.

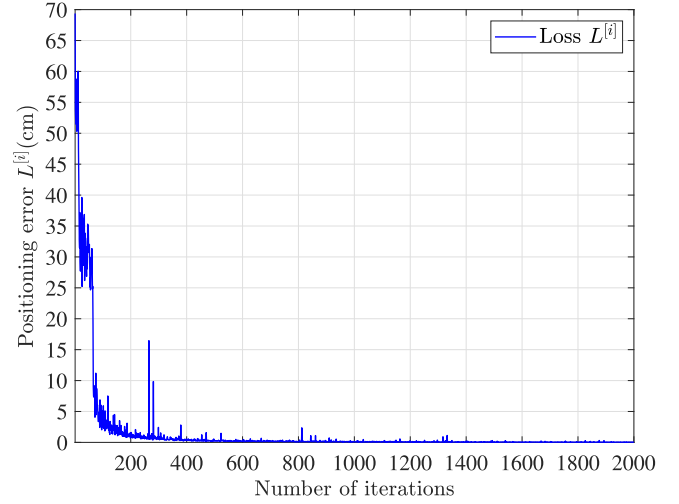
Fig. 10 compares positioning errors of SGD-AO and LS-AO schemes which includes 50 random test points. It is observed that, both the mean and variance of the positioning errors of SGD-AO are significantly lower than those of LS-AO schemes.

Fig. 11 illustrates the 3-D positioning results of SGD-AO scheme. In Fig. 11, the red dots denote the exact location coordinates and the blue dots denote corresponding positioning results, which verifies the positioning accuracy of the SGD-AO VLP scheme.

Table III compares the computational time, positioning error, number of layers, input dimension, number of nodes, and required training data set size of LS-VU, LS-AO, SGD-VU, and SGD-AO 3-D VLP schemes. From Table III, we can find that the positioning time consuming of LS-VU, LS-AO, SGD-VU and SGD-AO 3-D VLP schemes are, respectively, 680, 1200, 0.32, and 0.45 ms, where the positioning time consumption of SGD-VU 3-D VLP scheme is the shortest. The positioning time of SGD-VU and SGD-AO VLP schemes



(a)



(b)

Fig. 8. (a) Positioning error of SGD-VU network versus number of iterations. (b) Positioning error of SGD-AO network versus number of iterations.

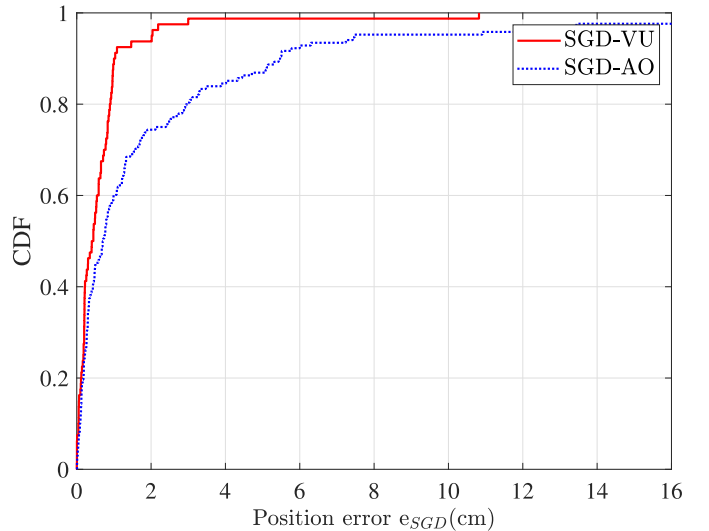


Fig. 9. Positioning errors CDF of SGD-VU and SGD-AO schemes.

are significantly lower than those of LS-VU and LS-AO VLP schemes. The reason is that LS-VU and LS-AO VLP schemes employ the SQP Algorithm to iteratively solve the positioning

TABLE III
PERFORMANCE COMPARISON AMONG THE PROPOSED SCHEMES

Schemes	Computational time	Positioning error	Number of layers	Input Dimension	Number of nodes	Dataset size
LS-VU	680ms	16.8cm	—	4	—	—
LS-AO	1200ms	26.36cm	—	7	—	—
SGD-VU	0.32ms	0.7cm	8	4	148	500
SGD-AO	0.45ms	1.77cm	14	7	395	2000

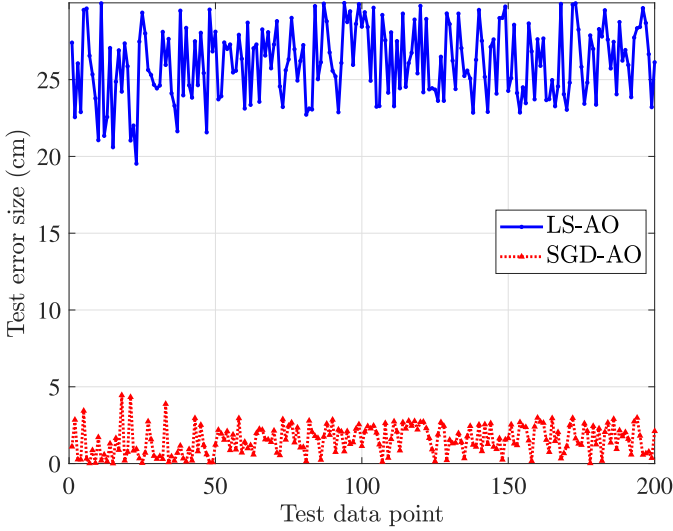


Fig. 10. Positioning performance comparison between LS-AO and SGD-AO schemes.

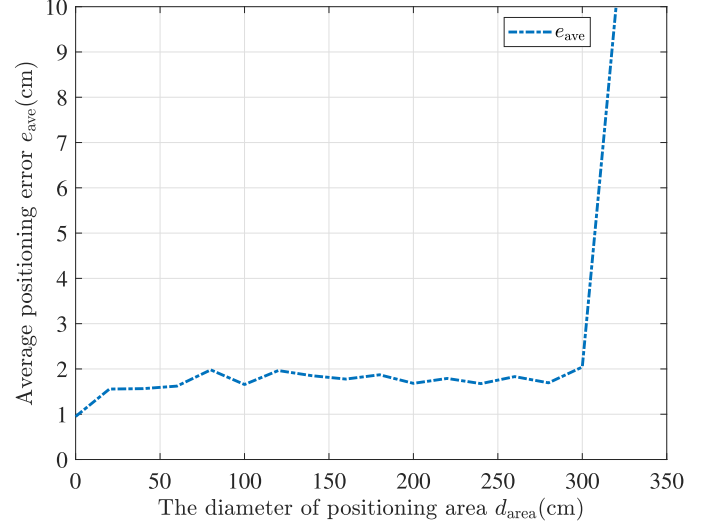


Fig. 12. Positioning error e_{ave} of the proposed SGD-AO network versus positioning area diameter d_{area} .

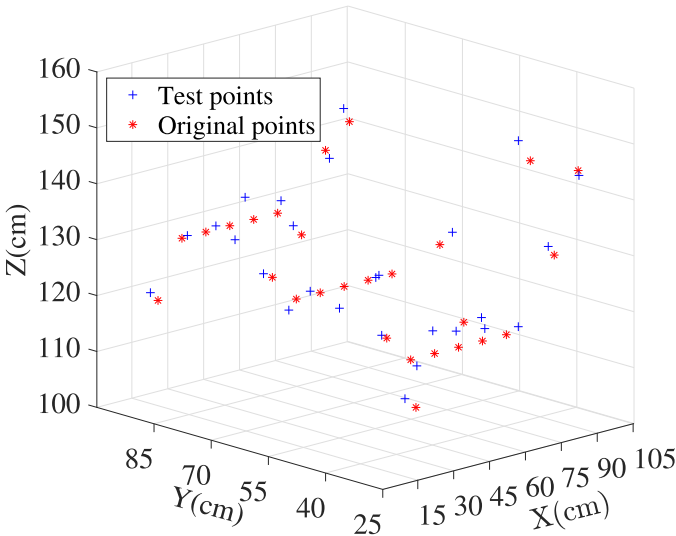


Fig. 11. Illustration of 3-D positioning performance of SGD-AO Scheme.

result, while SGD-VU and SGD-AO VLP schemes calculate the position through the pretrained networks. Table III also shows that the average positioning errors of the four schemes. The average positioning errors of SGD-VU and SGD-AO VLP schemes are significantly lower than those of LS-VU and LS-AO VLP schemes, which is consistent with our above intuition that the SQP algorithm may fall into some local optimum results. Comparing with the SQP

algorithm, deep learning-based SGD-VU and SGD-AO 3-D VLP schemes algorithms can handle complex nonlinear and nonconvex optimization problems.

Fig. 12 shows the positioning error e_{ave} of the proposed SGD-AO network versus positioning area diameter d_{area} . It can be seen that the positioning error is less than 2 cm for $d_{area} \leq 3$ m. However, for $d_{area} > 3$ m, the positioning error increases rapidly. The reason is that it exceeds the effective coverage of the single LED VLP system, and some PDs cannot receive light.

In the proposed VLP hardware platform, the range of the IMU angle error e_{IMU} is $e_{IMU} \in [-0.1^\circ, 0.1^\circ]$. To illustrate the effect of IMU angle error e_{IMU} on the proposed VLP schemes, we test the CDFs of the positioning accuracy with different e_{IMU} , as shown in Fig. 13.

Fig. 13 shows positioning accuracy CDF of different IMU angle error e_{IMU} . It can be seen that the positioning accuracy CDFs of $e_{IMU} \in [-0.1^\circ, 0.1^\circ]$ and $e_{IMU} \in [-10^\circ, 10^\circ]$ are close, which shows that the proposed VLP schemes are robust to IMU angle errors, especially for error $e_{IMU} \in [-10^\circ, 10^\circ]$. Moreover, for the large IMU angle error $e_{IMU} \in [-20^\circ, 20^\circ]$, the positioning performance degrades.

In addition, in order to illustrate the effect of the BNF algorithm, we compare the positioning performance of applying the BNF algorithm and without applying the BNF algorithm. The background noise of the positioning system includes both thermal noise and ambient light noise. Fig. 14 shows the CDF

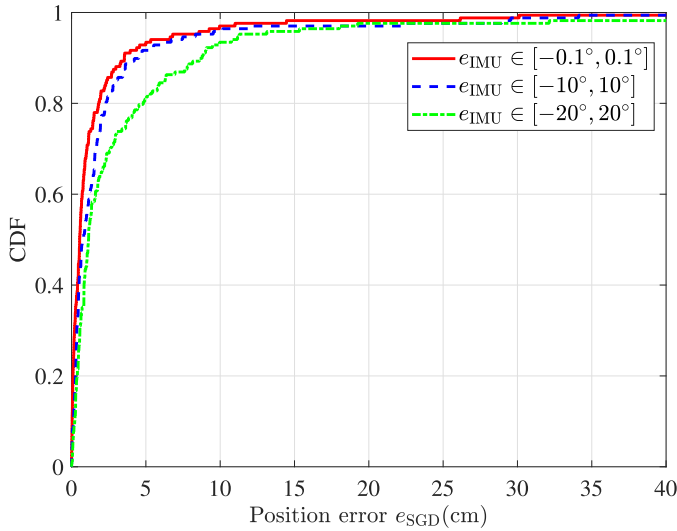


Fig. 13. Positioning accuracy CDF of different IMU angle error e_{IMU} .

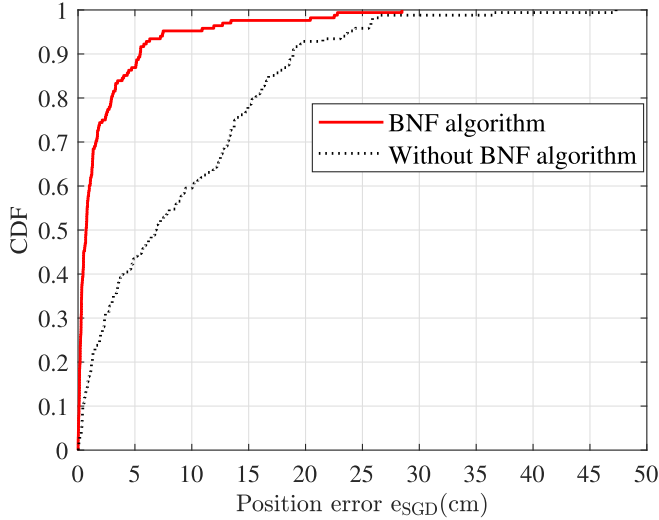


Fig. 14. CDF of the positioning error e_{SGD} of applying BNF algorithm and without applying BNF algorithm.

of the positioning error e_{SGD} of applying BNF algorithm and without applying BNF algorithm. It can be seen that by applying the BNF algorithm, the positioning error e_{SGD} can be significantly reduced, and the average positioning error is reduced from 8.84 to 1.77 cm.

VII. CONCLUSION

In this article, we proposed centimeter-level 3-D mobile online VLP schemes for the UE with the multiple PDs and one IMU. We first formulated the positioning problem as an NLS problem by considering arbitrary UE orientation, and obtained the UE location via the SQP positioning schemes. Then, in order to improve the accuracy, we turn to a deep learning method and developed SGD positioning networks. Moreover, we designed a 3-D single-lamp-multi-PDs positioning system prototype, where RaspberryPi 4 Model B is used as the positioning signal processor and data memory. Based on the designed VLP system prototype, we examined two positioning

schemes for the arbitrary UE orientation scenarios, and the corresponding positioning accuracy can reach 26.36 and 1.77 cm for LS-AO and SGD-AO schemes, respectively. Moreover, for the vertically up UE orientation scenarios, the developed LS-VU and SGD-VU positioning schemes can achieve 16.8- and 0.7-cm positioning accuracy, respectively. Under the same conditions, the positioning error of the proposed positioning schemes is significantly lower than that of the existing methods. In addition, we provided the first publicly available RSS and orientation measured database of 3-D VLP systems.

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